

OUSSAMA EL-ASRI

Full-Stack AI / GenAI Engineer · Software Engineering Student · Seeking 4–6 Month Internship

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SUMMARY

Self-directed Software Engineering student at **UM6P-1337 (42 Network)** specializing in full-stack AI and GenAI applications. Experienced in building RAG pipelines, LLM-powered agents, backend APIs, and containerized multi-service systems. Strong foundation in systems programming, networking, parsing, and software architecture. Highly autonomous learner accustomed to solving technical problems independently through documentation, research, and project-based development.

TECHNICAL SKILLS

Languages: Python, C, C++98, JavaScript, TypeScript

AI / GenAI: RAG, LangChain, Ollama, AI Agents, vector embeddings, prompt engineering, LangGraph, CrewAI

Frontend: React, Next.js, Tailwind CSS, Vite

Backend: FastAPI, Flask, Django, Express.js, REST APIs, Gunicorn, Uvicorn

Databases: PostgreSQL, SQLite, Redis, Prisma ORM, SQLAlchemy

DevOps: Docker, Docker Compose, NGINX, Git, Linux, uv

Systems: HTTP/1.1, TCP/IP, CGI, non-blocking I/O, concurrency, mutexes/semaphores, process management

CS Fundamentals: Data structures, parsing algorithms, recursive descent, AST construction, API design

SELECTED PROJECTS

AI Studio — Unified AI Platform

Personal · React, TypeScript, Vite, Python, Flask, FastAPI, LangChain, Ollama, Docker Compose

- Built a full-stack AI platform integrating two independent AI services behind a unified React interface.
- Developed a **RAG assistant** that ingests arbitrary URLs, chunks content, generates vector embeddings with `nomic-embed-text`, and performs semantic retrieval for contextual Q&A.
- Built a **Crypto Analysis Agent** using LangChain and LangGraph to process natural-language requests, retrieve 30-day OHLCV data from Binance, and calculate EMA, RSI, OBV, standard deviation, and Bollinger Bands.
- Designed a multi-container architecture with separate Flask/FastAPI services and a shared local Ollama LLM runtime.

ft_transcendence — Real-Time Multiplayer Platform

Team project · Next.js, Express.js, Socket.io, Prisma, SQLite, Python, Flask, LangChain, Ollama, Docker

- Developed the **AI subsystem** of a full-stack real-time multiplayer platform as part of a 4-person team.
- Designed and implemented the RAG-powered AI assistant using LangChain and Ollama, including content retrieval, embeddings, prompt orchestration, and backend integration.
- Integrated the AI service into a broader platform featuring real-time multiplayer gaming, WebSocket chat, JWT/OAuth authentication, 2FA, statistics, and analytics.
- Worked within a modular multi-service architecture orchestrated with Docker Compose and NGINX.

Webserv — HTTP/1.1 Web Server

Team project · C++98, `poll()`, CGI, Python, PHP-CGI

- Implemented HTTP response generation for **GET, POST, and DELETE**, including status codes, headers, payloads, file uploads, and error responses.
- Developed Python CGI execution, including process management, environment setup, request-body handling, and CGI output parsing.
- Contributed to a non-blocking HTTP/1.1 server architecture using a single `poll()` event loop for concurrent connections.

Weather App — Full-Stack Weather Platform

Personal · React, Tailwind CSS, FastAPI, SQLAlchemy, PostgreSQL

- Built a full-stack weather application with a React frontend and FastAPI backend.
- Designed REST APIs and PostgreSQL data models using SQLAlchemy.
- Developed the application independently from backend architecture through frontend integration.

Additional Projects

JSON Parser — Python recursive-descent JSON parser with custom lexer, AST construction, strict parsing, error positioning, and AST-to-Python evaluation.

Minishell — C Unix shell implementing command execution, pipelines, redirections, built-ins, parsing, and process management.

Inception — Docker Compose infrastructure running NGINX, WordPress/PHP-FPM, and MariaDB with TLS, networking, and persistent storage.

EDUCATION

UM6P-1337 — 42 Network, Morocco · 2023–Present

Software Engineering Student · Expected Bachelor's completion after internship

Project-based, peer-to-peer curriculum covering software engineering, algorithms, systems programming, networking, DevOps, and AI.

FST Al Hoceima · 2020–2022

MIP — Mathematics, Computer Science & Physics

Baccalaureate — Physical Sciences · 2019

CERTIFICATIONS & ADDITIONAL

AI / Software: Anthropic — AI Fluency: Framework and Foundations · IBM — Applied Software Engineering Fundamentals · IBM — Python for Data Science, AI & Development · Vercel — React Foundations for Next.js · University of Leeds — Foundations of Software Testing and Validation

English: EF SET — 51/100 (B2 Upper-Intermediate) · **Languages:** Arabic — Native · English — B2 · French — Conversational

Competitive Programming: LeetCode · CodinGame · HackerRank